

XUANCHI REN

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EDUCATION

University of Toronto , Toronto, ON, Canada Ph.D. in Computer Science Advisor: Prof. Sanja Fidler	Sept. 2022 – Sept. 2026
The Hong Kong University of Science and Technology , Hong Kong B.Eng. in Computer Science and B.Sc. in General Mathematics Advisor: Prof. Qifeng Chen	Aug. 2017 – Jun. 2022
École Polytechnique Fédérale de Lausanne (EPFL) , Switzerland Computer Science Exchange Program Advisor: Prof. Alexandre Alahi	Feb. 2020 – Jun. 2020

EMPLOYMENT

NVIDIA Spatial Intelligence Lab , Toronto Senior Research Scientist	May 2026 – July 2026
NVIDIA Spatial Intelligence Lab , Toronto Research Scientist	Sep. 2025 – Apr. 2026
NVIDIA Spatial Intelligence Lab , Toronto Research Intern	Nov. 2022 – Sep. 2025
Microsoft Research Asia , Beijing Research Intern	Jun. 2020 – Dec. 2021

RESEARCH

* indicates equal contribution; † indicates joint advising

Preprints

- PiD: Fast and High-Resolution Latent Decoding with Pixel Diffusion**
Yifan Lu, Qi Wu, Jay Zhangjie Wu, Zian Wang, Huan Ling, Sanja Fidler, **Xuanchi Ren**
arXiv preprint, 2026, [Project](#), [Paper](#), [Code](#), [Model](#)
- Gamma-World: Generative Multi-Agent World Modeling Beyond Two Players**
Fangfu Liu*, Kai He*, Tianchang Shen, Tianshi Cao, Sanja Fidler, Yueqi Duan, Jun Gao, Igor Gilitschenski†, Zian Wang†, **Xuanchi Ren**†
arXiv preprint, 2026, [Project](#), [Paper](#), [Code](#)
- NVIDIA OmniDreams: Real-Time Generative World Model for Closed-Loop Autonomous Vehicle Simulation**
NVIDIA (**Core Contributor**; Model Training)
White Paper, [Project](#), [Paper](#), [Code](#)
- Cosmos 3: Omnimodal World Models for Physical AI**
NVIDIA (**Contributor**; Working on Cosmos3-Interleaved)
White Paper, [Project](#), [Paper](#), [Code](#)

5. **Cosmos-Drive-Dreams: Scalable Synthetic Driving Data Generation with World Foundation Models**
Xuanchi Ren*, Yifan Lu*, Tianshi Cao*, Ruiyuan Gao*, Shengyu Huang, Amirmojtaba Sabour, Tianchang Shen, Tobias Pfaff, Jay Zhangjie Wu, Runjian Chen, Seung Wook Kim, Jun Gao, Laura Leal-Taixe, Mike Chen, Sanja Fidler, Huan Ling
 White Paper, [Project](#), [Paper](#), [Code](#)
6. **Cosmos-Transfer1: Conditional World Generation with Adaptive Multimodal Control**
 NVIDIA (**Core Contributor**; Featured in Jensen Huang’s GTC 2025 keynote)
 White Paper, [Project](#), [Paper](#), [Code](#)
7. **ViPE: Video Pose Engine for 3D Geometric Perception**
 Jiahui Huang, Qunjie Zhou, Hesam Rabeti, Aleksandr Korovko, Huan Ling, **Xuanchi Ren**, Tianchang Shen, Jun Gao, Dmitry Slepichev, Chen-Hsuan Lin, Jiawei Ren, Kevin Xie, Joydeep Biswas, Laura Leal-Taixe, Sanja Fidler
 White Paper, [Project](#), [Paper](#), [Code](#)
8. **World Simulation with Video Foundation Models for Physical AI**
 NVIDIA (**Contributor**)
 White Paper, [Project](#), [Paper](#), [Code](#)

Conference Papers

1. **Lyra 2.0: Explorable Generative 3D Worlds**
 Tianchang Shen*, Sherwin Bahmani, Kai He, Sangeetha Grama Srinivasan, Tianshi Cao, Jiawei Ren, Ruilong Li, Ruilong Li, Zian Wang, Nicholas Sharp, Zan Gojcic, Sanja Fidler, Jiahui Huang, Huan Ling, Jun Gao, **Xuanchi Ren***
 SIGGRAPH Asia, 2026, [Project](#), [Paper](#), [Code](#)
2. **MoRight: Motion Control Done Right**
 Shaowei Liu, **Xuanchi Ren**, Tianchang Shen, Huan Ling, Saurabh Gupta, Shenlong Wang, Sanja Fidler, Jun Gao
 European Conference on Computer Vision (ECCV), 2026, [Project](#), [Paper](#)
3. **Lyra: Generative 3D Scene Reconstruction via Video Diffusion Model Self-Distillation**
 Sherwin Bahmani, Tianchang Shen, Jiawei Ren, Jiahui Huang, Yifeng Jiang, Haithem Turki, Andrea Tagliasacchi, David B. Lindell, Zan Gojcic, Sanja Fidler, Huan Ling, Jun Gao*, **Xuanchi Ren***
 International Conference on Learning Representations (ICLR), 2026, [Project](#), [Paper](#), [Code](#)
4. **ChronoEdit: Towards Temporal Reasoning for Image Editing and World Simulation**
 Jay Zhangjie Wu*, **Xuanchi Ren***, Tianchang Shen, Tianshi Cao, Kai He, Yifan Lu, Ruiyuan Gao, Enze Xie, Shiyi Lan, Jose M. Alvarez, Jun Gao, Sanja Fidler, Zian Wang, Huan Ling
 International Conference on Learning Representations (ICLR), 2026, [Project](#), [Paper](#), [Code](#)
5. **InfiniCube: Unbounded and Controllable Dynamic 3D Driving Scene Generation with World-Guided Video Models**
 Yifan Lu*, **Xuanchi Ren***, Jiawei Yang, Tianchang Shen, Jay Zhangjie Wu, Jun Gao, Yue Wang, Siheng Chen, Mike Chen, Sanja Fidler, Jiahui Huang
 International Conference on Computer Vision (ICCV), 2025, [Project](#), [Paper](#), [Code](#)
6. **GEN3C: 3D-Informed World-Consistent Video Generation with Precise Camera Control**
Xuanchi Ren*, Tianchang Shen*, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas Müller, Alexander Keller, Sanja Fidler, Jun Gao
 Computer Vision and Pattern Recognition (CVPR), 2025, (**Highlight**), [Project](#), [Paper](#), [Code](#)

7. **Difix3D+: Improving 3D Reconstructions with Single-Step Diffusion Models**
Jay Zhangjie Wu*, Yuxuan Zhang*, Haithem Turki, **Xuanchi Ren**, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic†, Huan Ling†
Computer Vision and Pattern Recognition (CVPR), 2025, (**Oral, Best Paper Award Candidate**), [Project](#), [Paper](#), [Code](#)
8. **SCube: Instant Large-Scale Scene Reconstruction using VoxSplats**
Xuanchi Ren*, Yifan Lu*, Hanxue Liang, Jay Zhangjie Wu, Huan Ling, Mike Chen, Sanja Fidler, Francis Williams, Jiahui Huang
Neural Information Processing Systems (NeurIPS), 2024, [Project](#), [Paper](#), [Code](#)
9. **XCube: Large-Scale 3D Generative Modeling using Sparse Voxel Hierarchies**
Xuanchi Ren, Jiahui Huang, Xiaohui Zeng, Ken Museth, Sanja Fidler, Francis Williams
Computer Vision and Pattern Recognition (CVPR), 2024, (**Highlight**), [Project](#), [Paper](#), [Code](#)
10. **Blind Video Deflickering by Neural Filtering with a Flawed Atlas**
Chenyang Lei*, **Xuanchi Ren***, Zhaoxiang Zhang, Qifeng Chen
Computer Vision and Pattern Recognition (CVPR), 2023, [Project](#), [Paper](#), [Code](#)
11. **Look Outside the Room: Synthesizing A Consistent Long-Term 3D Scene Video from A Single Image**
Xuanchi Ren, Xiaolong Wang
Computer Vision and Pattern Recognition (CVPR), 2022, [Project](#), [Paper](#), [Code](#)
12. **Learning Disentangled Representation by Exploiting Pretrained Generative Models: A Contrastive Learning View**
Xuanchi Ren*, Tao Yang*, Yuwang Wang, Wenjun Zeng
International Conference on Learning Representations (ICLR), 2022, [Paper](#), [Code](#)
13. **Retriever: Learning Content-Style Representation as a Token-Level Bipartite Graph**
Dacheng Yin*, **Xuanchi Ren***, Chong Luo, Yuwang Wang, Zhiwei Xiong, Wenjun Zeng
International Conference on Learning Representations (ICLR), 2022, [Project](#), [Paper](#), [Code](#)
14. **Towards Building A Group-based Unsupervised Representation Disentanglement Framework**
Tao Yang, **Xuanchi Ren**, Yuwang Wang, Wenjun Zeng, Nanning Zheng
International Conference on Learning Representations (ICLR), 2022, [Paper](#)
15. **Safety-Aware Motion Prediction with Unseen Vehicles for Autonomous Driving**
Xuanchi Ren*, Tao Yang*, Li Erran Li, Alexandre Alahi, Qifeng Chen
International Conference on Computer Vision (ICCV), 2021, [Paper](#), [Code](#)
16. **Self-Supervised Dance Video Synthesis Conditioned on Music**
Xuanchi Ren, Haoran Li, Zijian Huang, Qifeng Chen
ACM International Conference on Multimedia (ACM MM), 2020, (**Oral**), [Paper](#), [Code](#)

Journal Papers

1. **fVDB: A Deep-Learning Framework for Sparse, Large-Scale, and High-Performance Spatial Intelligence**
Francis Williams, Jiahui Huang, Jonathan Swartz, Gergely Klár, Vijay Thakkar, Matthew Cong, **Xuanchi Ren**, Ruilong Li, Clement Fuji-Tsang, Sanja Fidler, Eftychios Sifakis, Ken Museth
ACM Transactions on Graphics (TOG), 2024, [Project](#), [Paper](#)

HONORS & AWARDS

- Best Paper Award Candidate at CVPR 2025

- Vector Scholarships in Artificial Intelligence, Vector Institute 2022
- Hong Kong PhD Fellowship Scheme 2022

ACADEMIC SERVICES

Conference Reviewer

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV)
- Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)

Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
- IEEE Transactions on Multimedia (TMM)
- ACM SIGGRAPH
- ACM SIGGRAPH Asia